

CAD Viewer — User Manual

CAD Viewer — User Manual

Version: v0.2 · **URL:** <http://localhost:4322/>

CAD Viewer is a real-time collaborative browser app for viewing CAD geometry (GLBs from SolidEdge, FreeCAD, Fusion 360, OpenCascade..., plus STEP files converted on the server). Create a session, share a model, and everyone on your LAN follows your camera and part visibility live — or just use it as a plain local viewer.

1. Starting the app

Portable zip (Windows, no installs):

1. Unzip `cad-viewer-portable.zip` anywhere.
2. Double-click `start.bat`. It launches the server and opens the browser.
3. The app is served at **`http://localhost:4322/`**

From source: `node server.js` in the project folder, then open the same URL.

Changing the port

The default port is **4322**. Any of these overrides it:

Method	Example
Launcher argument (zip)	<code>start.bat 5555</code>
<code>port.txt</code> file next to <code>server.js</code>	create <code>port.txt</code> containing <code>5555</code>
Environment variable	<code>set PORT=5555</code> then <code>node server.js</code> (Windows) / <code>PORT=5555 node server.js</code> (macOS/Linux)

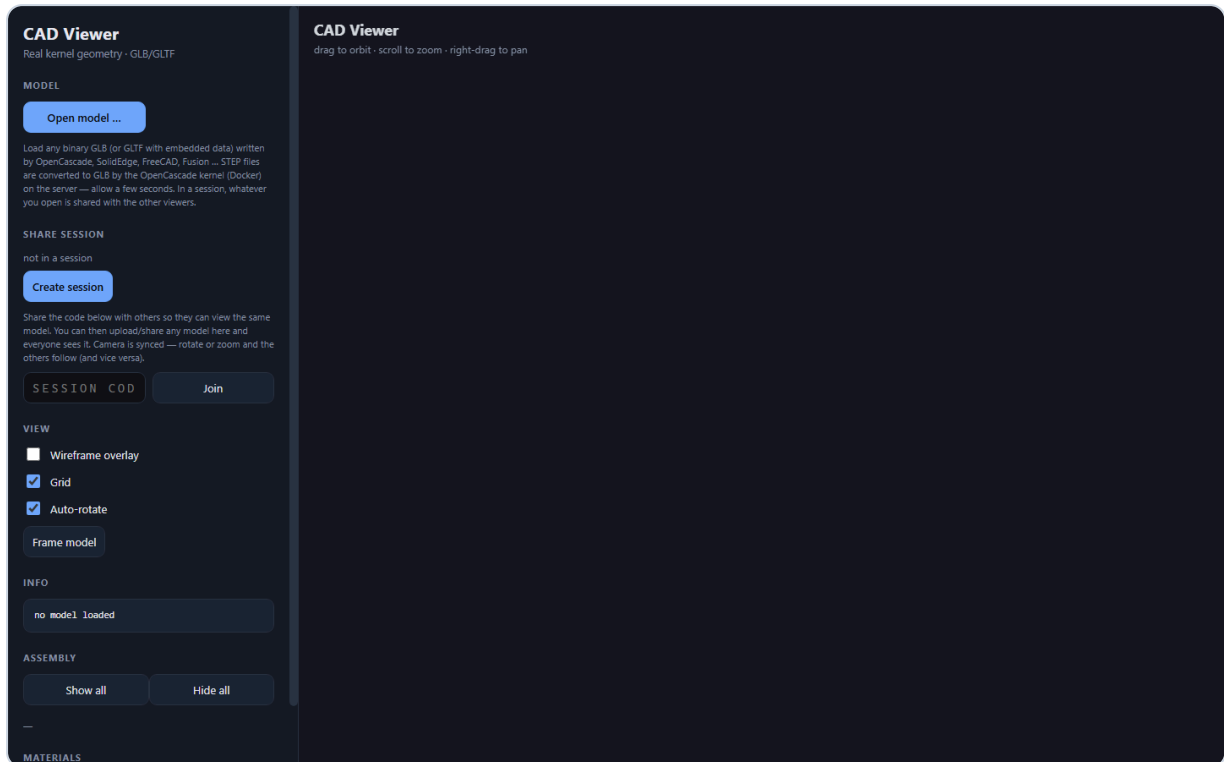
Precedence: command-line argument → `port.txt` → env `PORT` → default 4322.

The join link and `/ip` address always reflect the actual port, so guests don't need to know it.

STEP files (.step/.stp) additionally need Docker + the `chair-cq:local`

OpenCascade image. Run `install-docker-opencascade.bat` once to set that

up. GLB/GLTF files work without any of it.

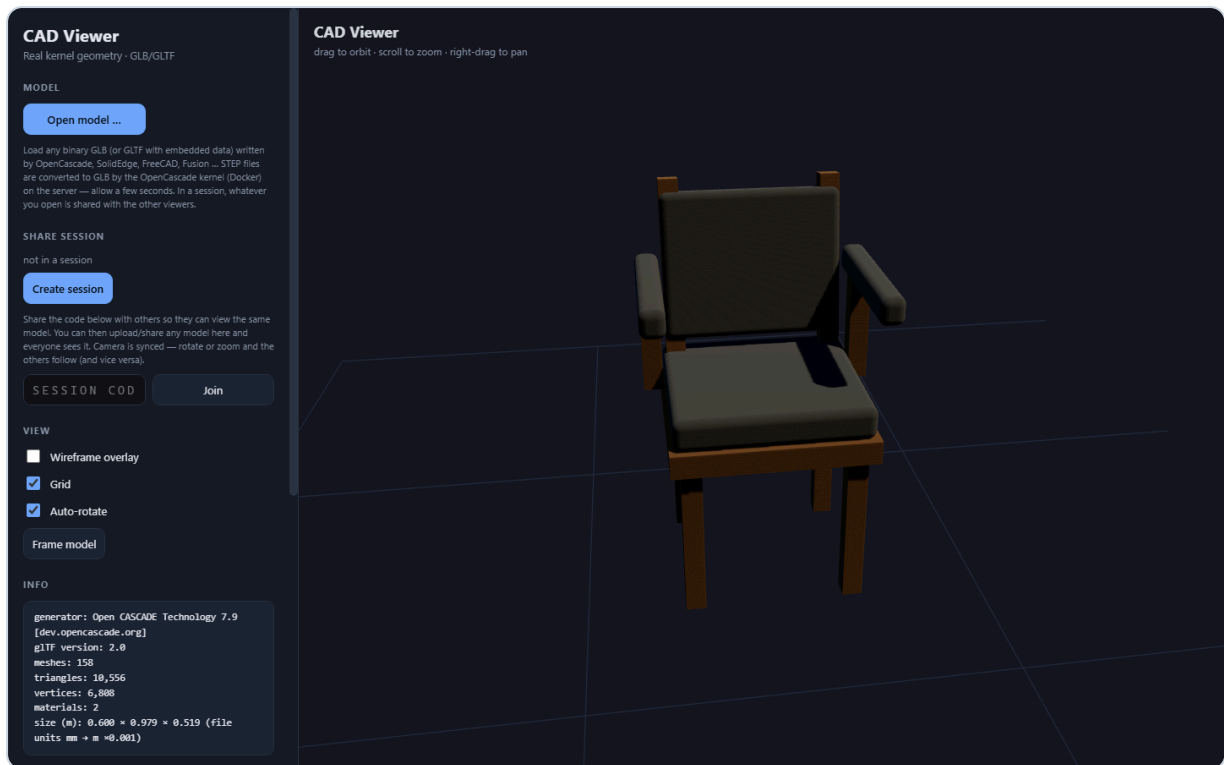


Empty start — the viewport is blank until you open a model

2. Opening a model

1. Click **Open model...** in the left sidebar.
2. Choose a `.glb`, `.gltf`, `.step` or `.stp` file from your computer.

GLB/GLTF loads instantly. STEP files are converted to GLB by the OpenCascade kernel on the server (a blocking overlay shows progress — a few seconds).



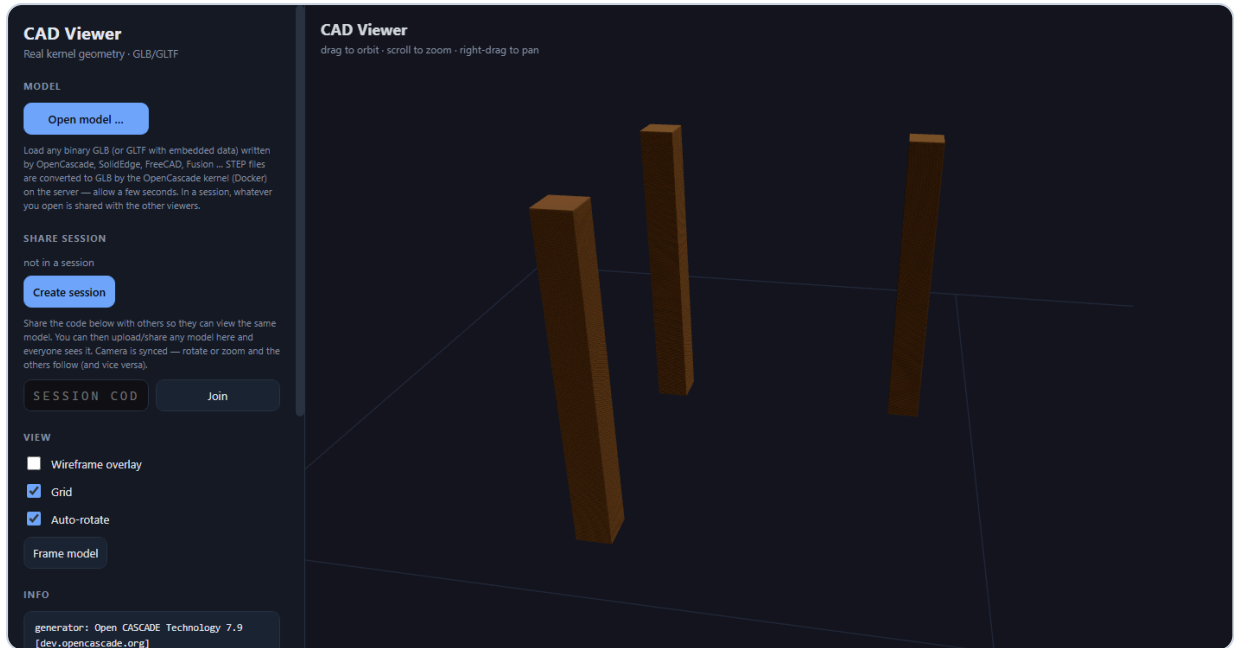
A model loaded in the viewport, with Info, Assembly and Materials panels

The sidebar shows you:

- **Info** — generator, mesh/triangle counts, dimensions and units
- **Assembly** — the part tree (see §3)
- **Materials** — swatches of every colour in the model

3. The Assembly panel — parts and visibility

Every part of the model appears in the **Assembly** tree. The tree opens expanded to **level 2** (the top assembly plus its direct children); deeper levels are collapsed until you expand them.



Assembly tree with the part list

Expand / collapse

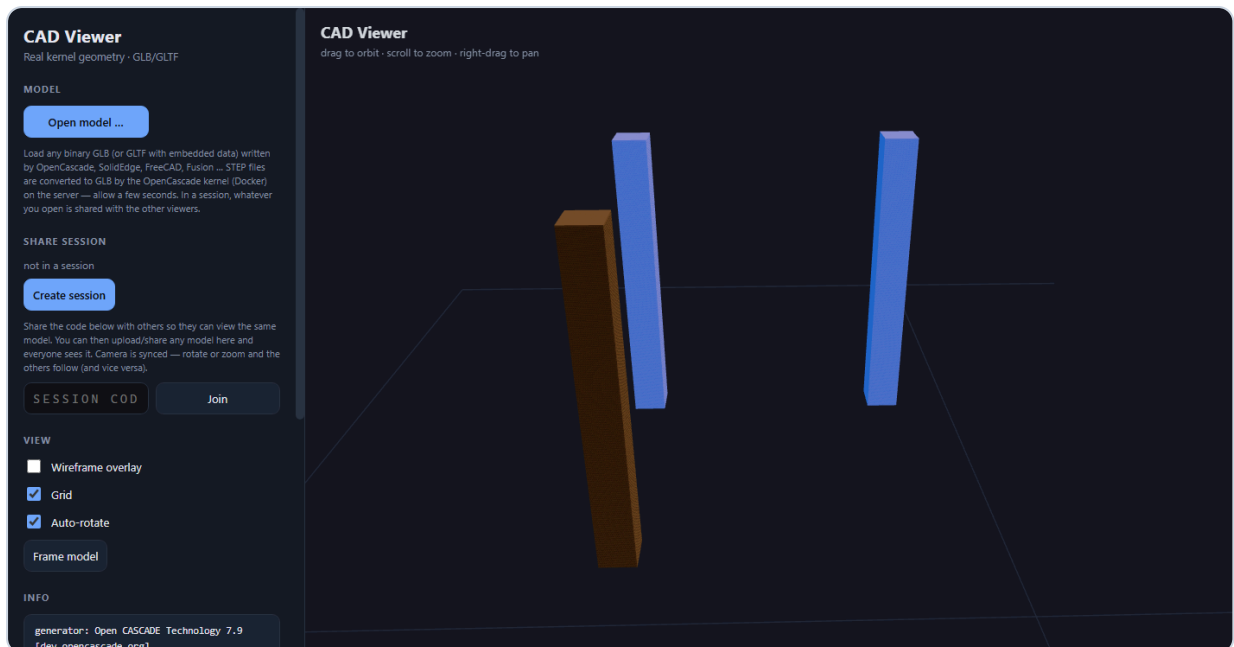
- Rows that contain sub-parts show a **+/- toggle** on the left.
- Click **+** to expand (show children), **-** to collapse (hide them).
- Expanding/collapsing is purely visual — it never changes part visibility.

Show / hide parts

- **Checkbox** per part: uncheck to hide it in the viewport, check to show.
- Toggling a parent cascades to its whole subtree — turning a sub-assembly on brings all its children back with it (three.js needs the whole chain visible to render anything).
- **Show all / Hide all** buttons apply to the whole model at once.

Highlight a part (click the name)

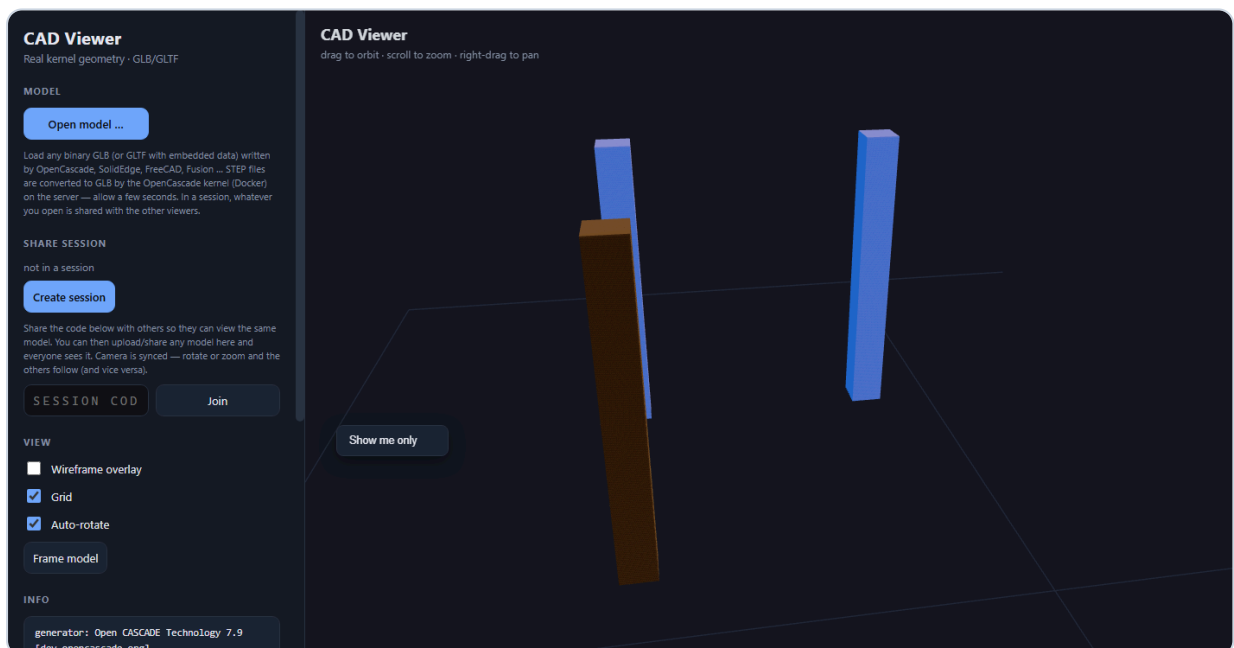
Click a part's **name** to highlight it in blue in the viewport. Click another part to move the highlight; click the same part again to clear it. The highlight uses per-mesh material copies, so other parts sharing the same source material are unaffected.



A part selected in the tree, highlighted blue in the viewport

"Show me only" (right-click)

Right-click a part and choose **Show me only** to hide everything except that part and its children. Ancestors stay visible so the isolated part still renders.



Right-click context menu with "Show me only"

In a session (see §5), all of this syncs to every member — show/hide, expand/collapse, and the selection highlight are mirrored live in both directions.

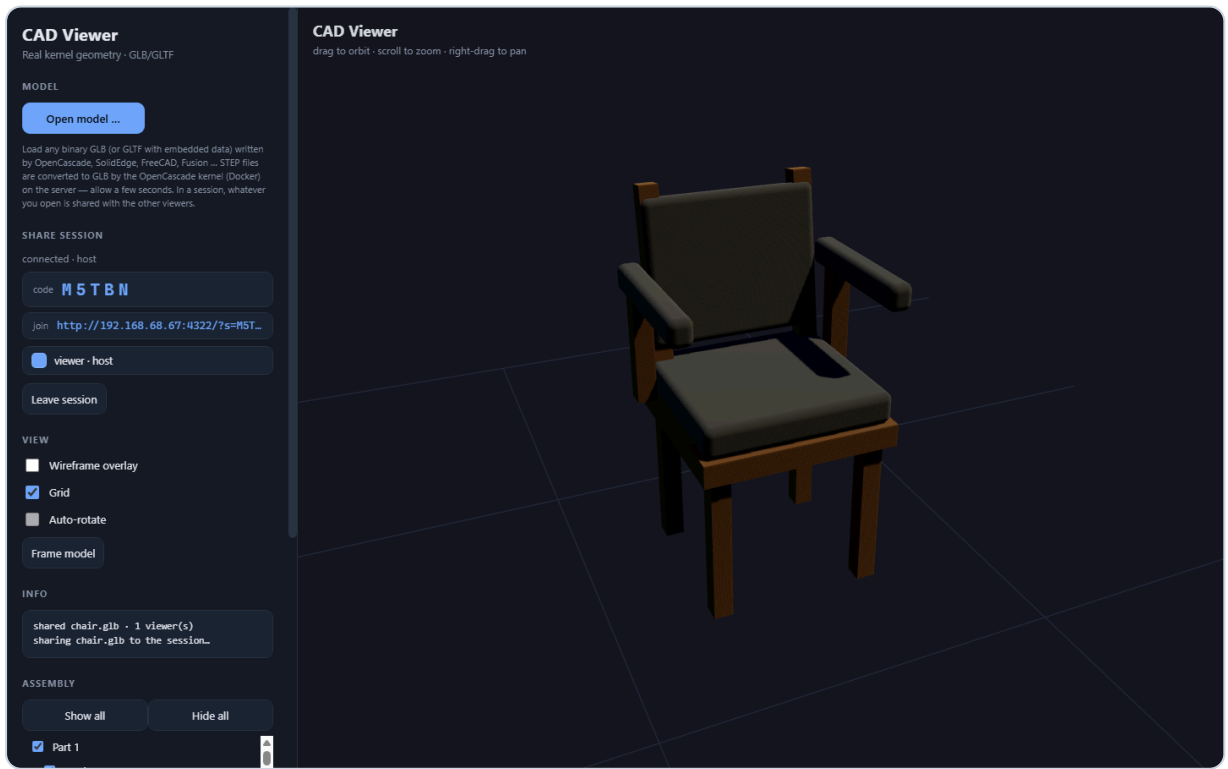
4. Navigating the viewport

Action	Effect
Drag (left button)	Orbit the camera
Scroll	Zoom
Right-drag	Pan
Frame model button	Reset the view to fit the model
Wireframe overlay	Toggle wireframe on all parts
Grid	Toggle the ground grid
Auto-rotate	Slow turntable spin (disabled in a session)

5. Collaborative sessions (host)

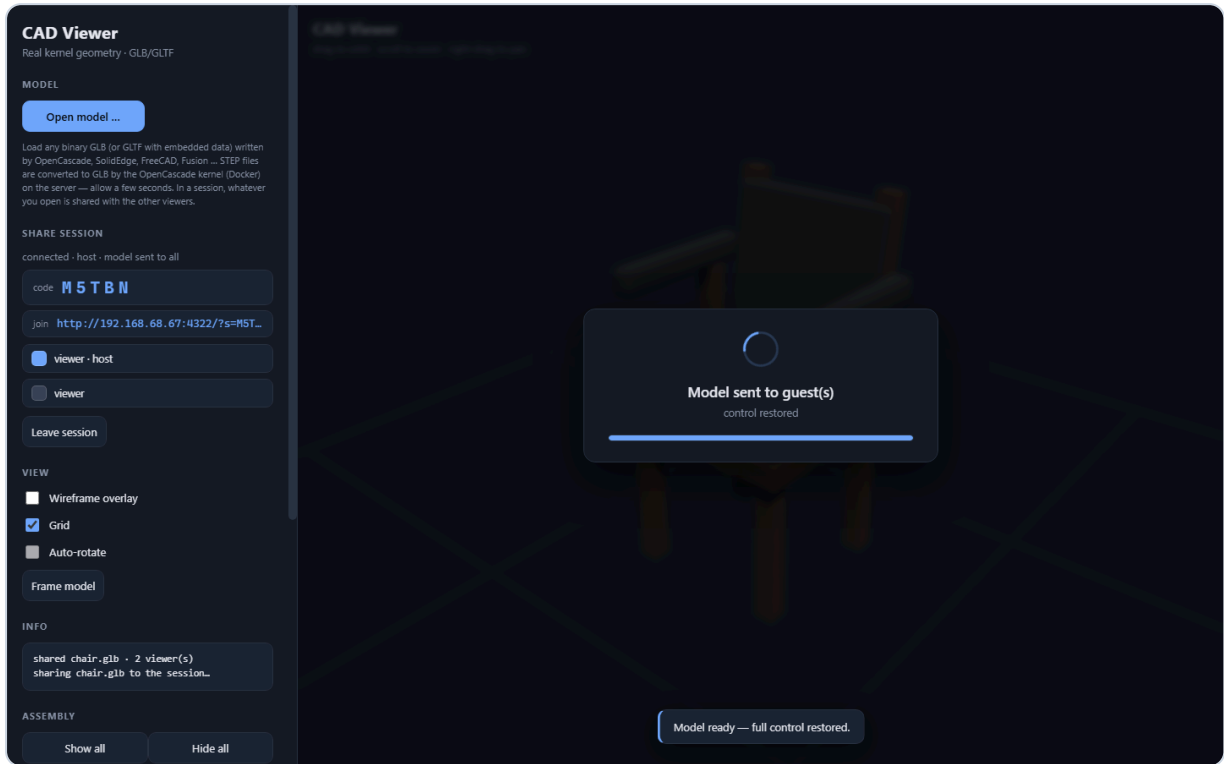
Sessions let other people on your LAN view the same model and follow your camera and part visibility.

1. Click **Create session** — a 5-character code appears.
2. **Open a model** (or re-share one you've already loaded).
3. Share the **join link** shown under the code, or the code itself, with the other users.



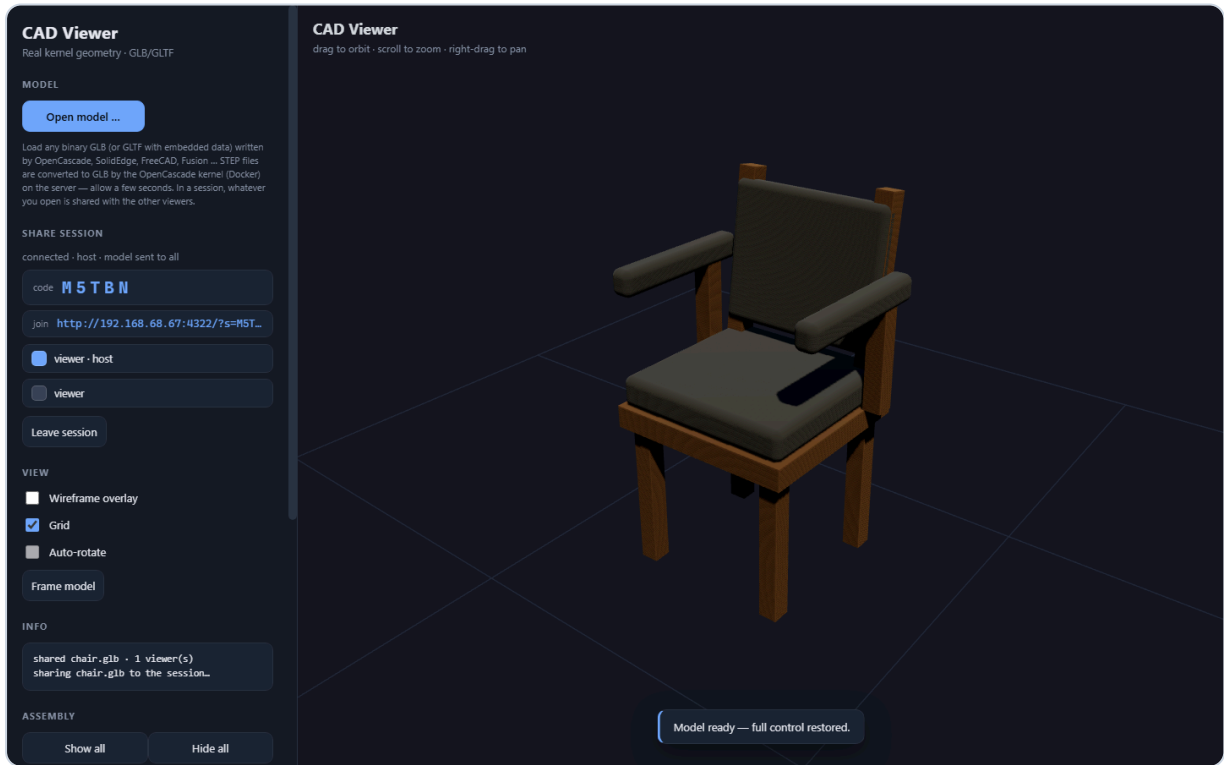
A session created — code and LAN join link shown

While the model is being handed to guests you'll see a blocking
“Sending model to guest(s)…” overlay; control is restored automatically once
every connected guest confirms receipt.



The host's “Sending model to guest(s)” overlay

The roster under the code shows who is connected.



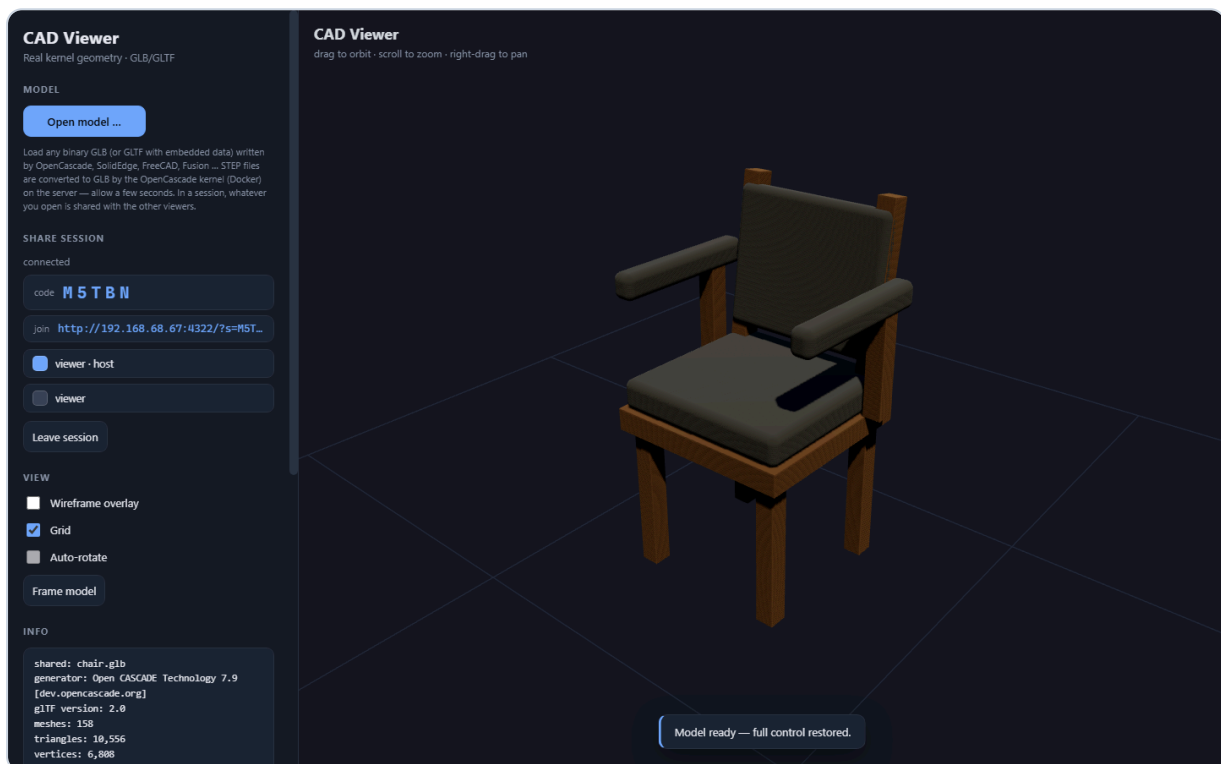
Host view: guest connected in the roster

6. Joining a session (guest)

On another computer (same network):

1. Open the **join link** the host sent you, **or**
2. Open the app, type the code into **session code** and click **Join**.

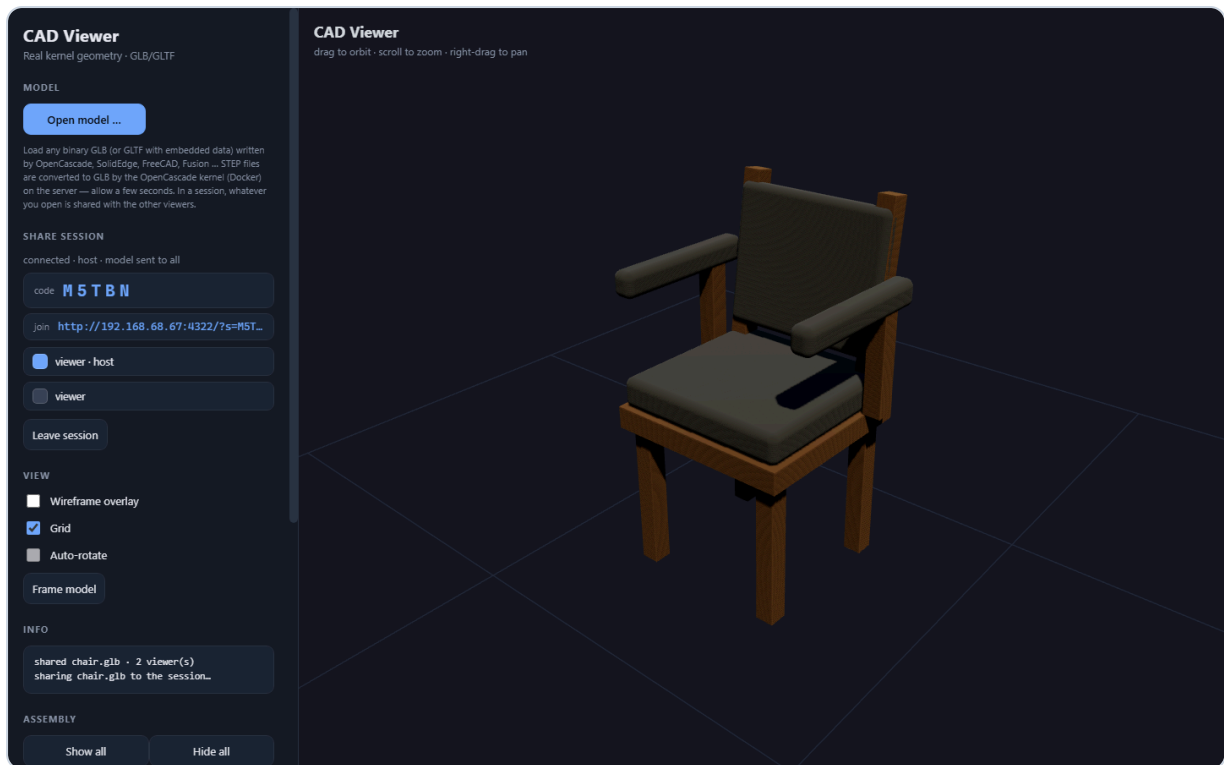
You'll see a **"Receiving model..."** overlay with byte progress, then the model appears — same view, same parts, same visibility as the host.



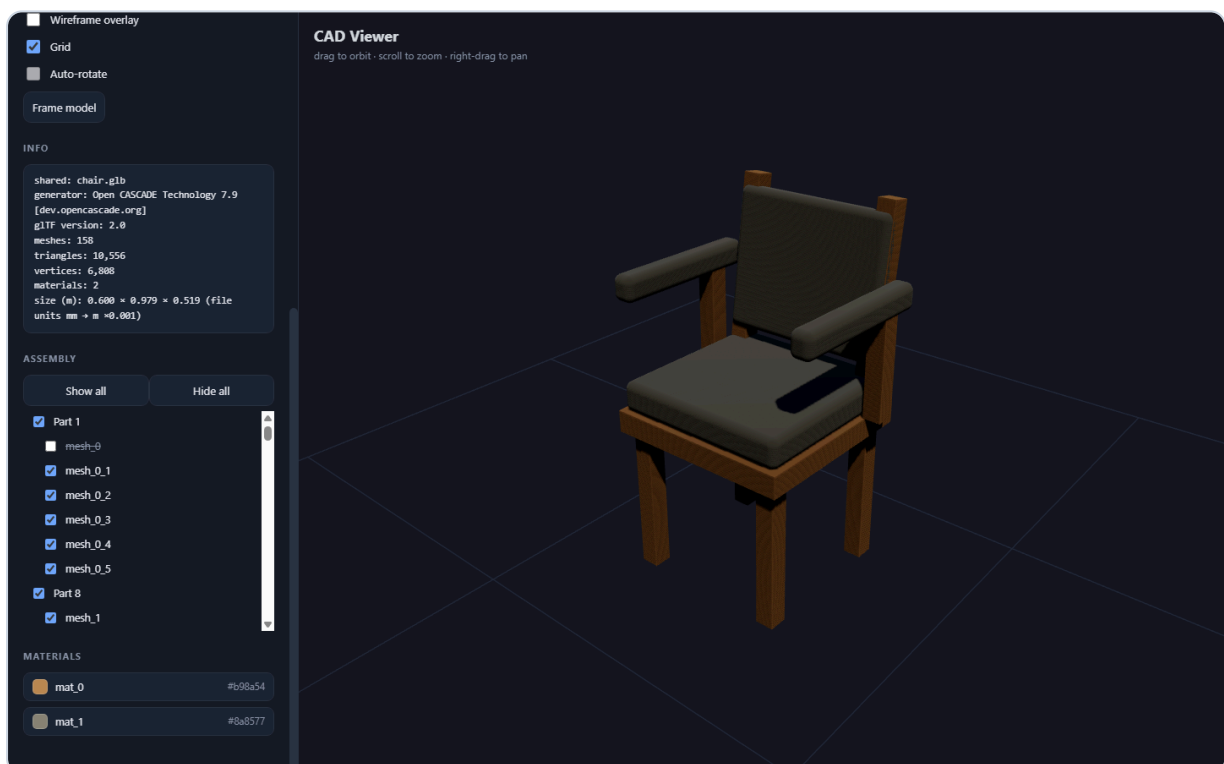
Guest view after receiving the shared model

What stays in sync

- **Camera** — whoever orbits/zooms/pans, everyone follows (both directions)
- **Part visibility** — show/hide any part and it changes for all members
- **The model itself** — late joiners automatically receive the current model and visibility state
- **Expand/collapse & selection** — the assembly-tree view and part highlight are mirrored live across members



Part hidden on the host

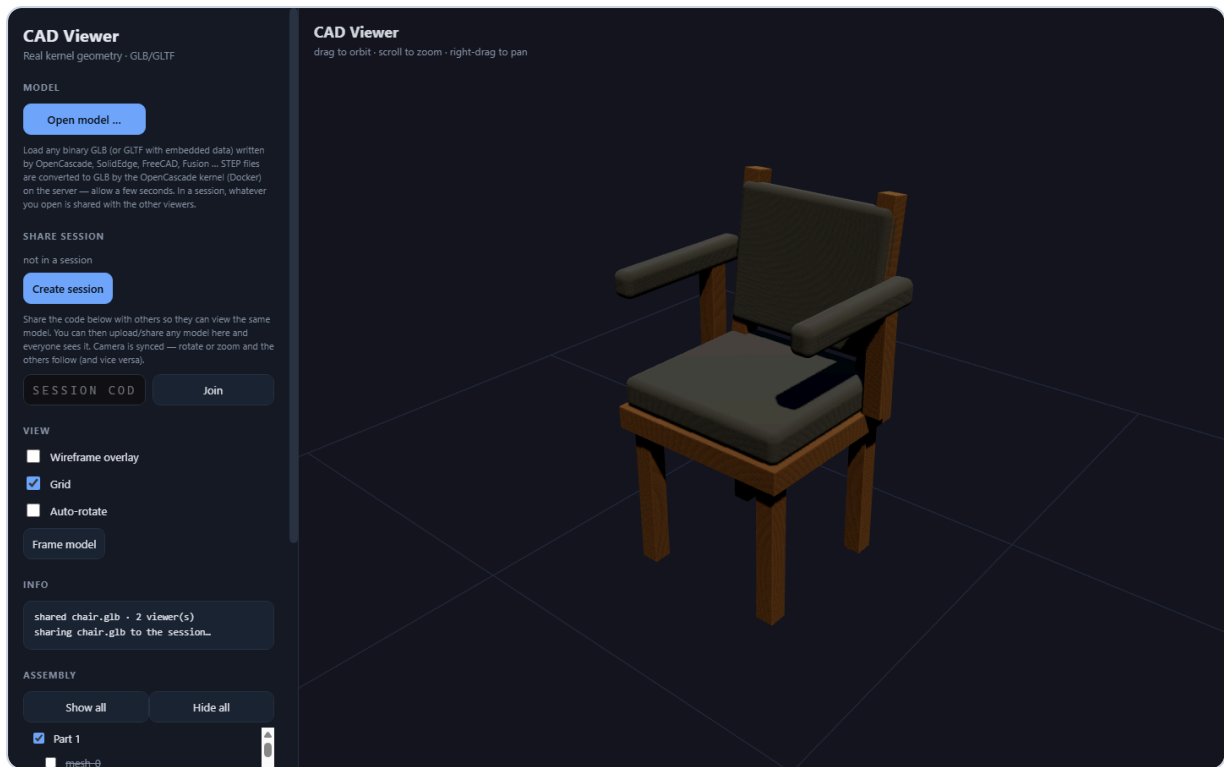


The same part hidden on the guest — synced

7. Leaving a session

Click **Leave session** — you keep your local model and return to solo viewing.

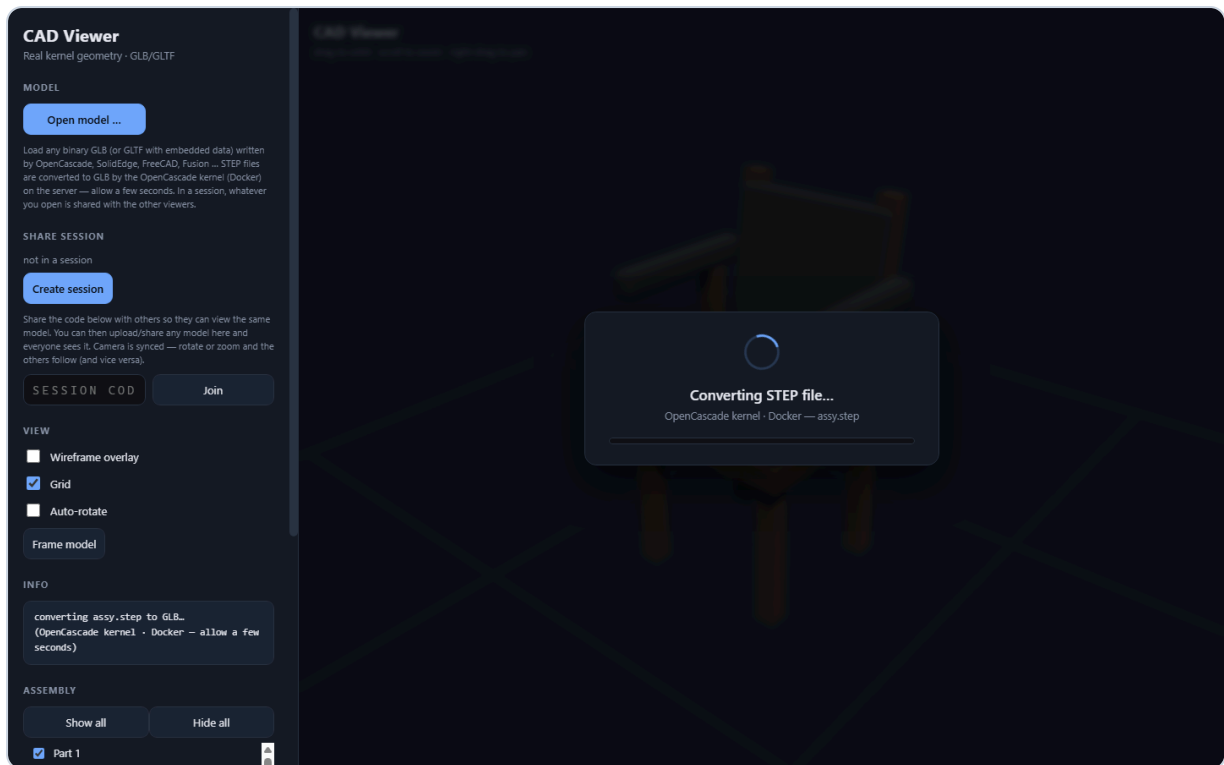
The session itself stays alive on the server for other members.



After leaving the session

8. Opening a STEP file

STEP conversion happens through the OpenCascade kernel (Docker). The first time you open a STEP file you'll see the conversion overlay; when it finishes the GLB is loaded — with the original **colours and materials** preserved, and part names appearing in the Assembly tree.



STEP file being converted to GLB

If Docker or the `chair-cq:local` image isn't installed, STEP files show a conversion error while GLB/GLTF continues to work normally.

9. Troubleshooting

Problem	Fix
Can't open the app on another PC	Use the join link (LAN address), make sure both PCs are on the same network, and that port 4322 isn't blocked by a firewall
STEP shows "conversion failed"	Run <code>install-docker-opencascade.bat</code> ; confirm Docker is running with the <code>chair-cq:local</code> image
Guest doesn't get the model	The host must be connected with a model loaded — joining an empty session shows nothing until the host shares
Port already in use	Change it: <code>start.bat 4323</code> (zip), a <code>port.txt</code> file, or <code>set PORT=4323</code>
Join link shows the wrong IP	The link uses the host's LAN address; refresh/re-create the session to re-detect it

CAD Viewer v0.22 — collaborative CAD viewing for the LAN.